The Discrete Wavelet Transform

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Key Concepts

1) The wavelet expansion is

$$x(t) = \sum_k c_{j_0}[k]\phi_{j_0,k}(t) + \sum_{j=j_0}^{J-1} \sum_k d_j[k]\psi_{j,k}(t)$$

2) The discrete wavelet transform (DWT) finds $c_{j_0}[k]$ and $d_j[k]$ with a computer, even though $x(t)$, $\phi(t)$ and $\psi(t)$ are continuous-time signals.

3) The key properties behind the DWT are orthonormality and the

scaling function and wavelet multi resolution recursions

$$\phi(t) = \sum_n h_0[n]\sqrt{2}\phi(2t-n)$$

$$\psi(t) = \sum_n h_1[n]\sqrt{2}\phi(2t-n)$$

4) The DWT computes $c_j[k]$ and $d_j[k]$ from $h_0[n]$, $h_1[n]$, and $c_J[k]$.

   a) The highest resolution scaling coefficients $c_J[k]$ are set equal to the samples of the signal.

   b) For $j = J-1$ down to j_0 compute

$$c_j[k] = \sum_m h_0[m-2k]c_{j+1}[m]$$

$$d_j[k] = \sum_m h_1[m - 2k] c_{j+1}[m]$$

5) The inverse DWT computes $c_{j+1}[k]$ from $h_0[n]$, $h_1[n]$, $c_j[k]$ and $d_j[k]$. For $j = j_0$ to $J - 1$ compute

$$c_{j+1}[k] = \sum_m c_j[m] h_0[k - 2m] + \sum_m d_j[m] h_1[k - 2m]$$

6) Both the DWT and the inverse DWT require $\mathrm{O}\big(N\big)$ numerical operations.

Video Screenshots

Wavelet expansion

$$x(t) = \sum_k c_{j_0}[k]\,\phi_{j_0,k}(t) + \sum_{j=j_0}^{J-1}\sum_k d_j[k]\,\psi_{j,k}(t)$$

resolution J

$c_{j_0}[k]$: scaling coefficients

$\phi_{j_0,k}(t)$: scaling function

$d_j[k]$: wavelet coefficients

$\psi_{j,k}(t)$: wavelet

$x(t)$, $\phi(t)$, $\psi(t)$ are continuous-time signals

How do we find $c_{j_0}[k]$, $d_j[k]$ with a computer?

Key facts –

$$c_j[k] = \langle \phi_{j,k}(t), x(t) \rangle = \int \phi_{j,k}(t)\,x(t)\,dt$$

$$d_j[k] = \langle \psi_{j,k}(t), x(t) \rangle = \int \psi_{j,k}(t)\,x(t)\,dt$$

o/n bases

$$\phi(t) = \sum_n h_0[n]\sqrt{2}\,\phi(2t-n)$$

$$\psi(t) = \sum_n h_1[n]\sqrt{2}\,\phi(2t-n)$$

Use $\phi_{j,k}(t) = 2^{j/2}\phi(2^j t - k)$, $\psi_{j,k}(t) = 2^{j/2}\psi(2^j t - k)$

$$\Rightarrow \quad \phi(2^j t - k) = \sum_m h_0[m-2k]\sqrt{2}\,\phi(2^{j+1}t - m)$$

$$\psi(2^j t - k) = \sum_m h_1[m-2k]\sqrt{2}\,\phi(2^{j+1}t - m)$$

Now

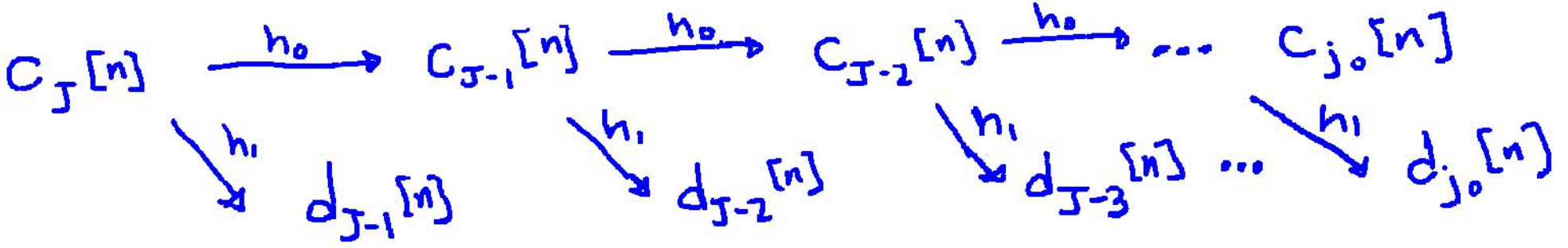
$$c_j[k] = \langle \phi_{j,k}(t), x(t)\rangle = \langle 2^{j/2}\phi(2^j t - k), x(t)\rangle$$

$$= \sum_m h_0[m-2k]\,\langle \underbrace{2^{(j+1)/2}\phi(2^{j+1}t - m)}_{\phi_{j+1,k}(t)}, x(t)\rangle$$

$$= \sum_m h_0[m-2k]\,\underbrace{\langle \cdots \rangle}_{c_{j+1}[m]}$$

$$c_j[k] = \sum_m h_0[m-2k]\,c_{j+1}[m]$$

$$d_j[k] = \sum_m h_1[m-2k]\,c_{j+1}[m]$$



Given $c_J[n]$, $h_0[n]$, $h_1[n]$ $\Rightarrow$ compute $c_j[n]$, $d_j[n]$, $j<J$

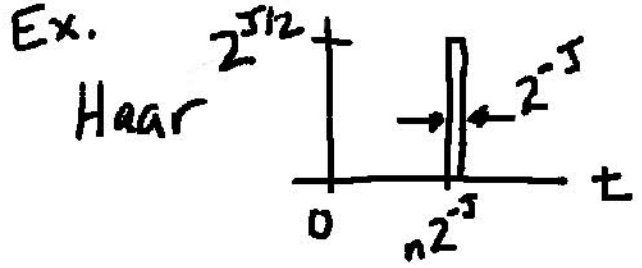
- How do we find $c_J[n]$?

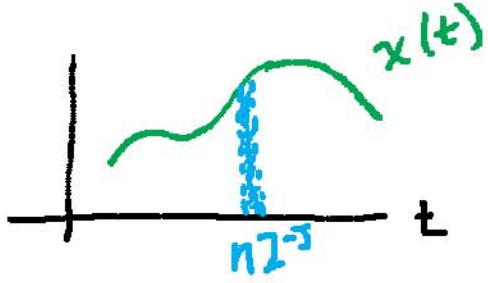
$$c_J[n] = \int \phi_{J,n}(t)\, x(t)\, dt$$

$$\sim x[n] = x(n2^{-J})$$

J large $\Rightarrow$ $\phi_{J,n}(t) \to c\delta(t-n)$

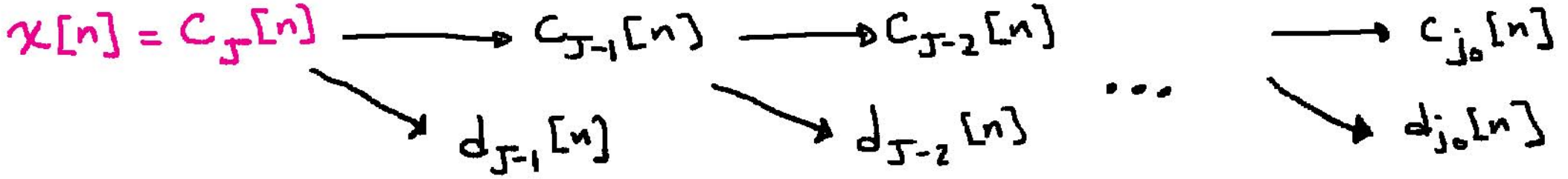
Ex. Haar





If $x(t)$ is sampled densely (sampling thm), $c_J[n] = x[n]$

$$c_j[k] = \sum_m h_0[m-2k]\, c_{j+1}[k], \quad d_j[k] = \sum_m h_1[m-2k]\, c_{j+1}[k]$$



Discrete Wavelet Transform (DWT)

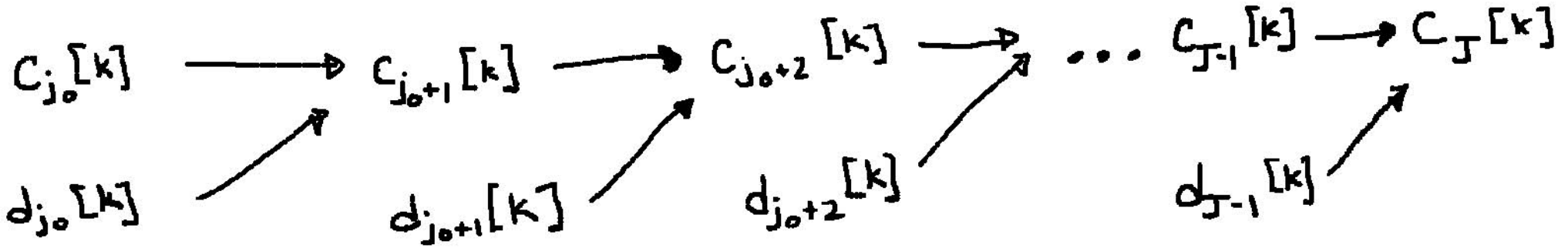
- If $x[n]$ has N values, $c_{J-1}, d_{J-1} \sim$ length $(N/2)$

  $\Rightarrow$ if $N = 2^L$, $j_0 \geq J-L$ (wavelet dependent)

- Computing DWT $\sim O(N)$ operations

Inverse DWT

$$c_{j+1}[k] = \sum_m c_j[m]\, h_0[k-2m] + \sum_m d_j[m]\, h_1[k-2m]$$



MATLAB

DWT

[W, P] = wavedec(x, Level, 'wname')

Level = 3

$$W = \left[\, c_{J-3} \;\vdots\; d_{J-3} \;\vdots\; d_{J-2} \;\vdots\; d_{J-1} \,\right]$$

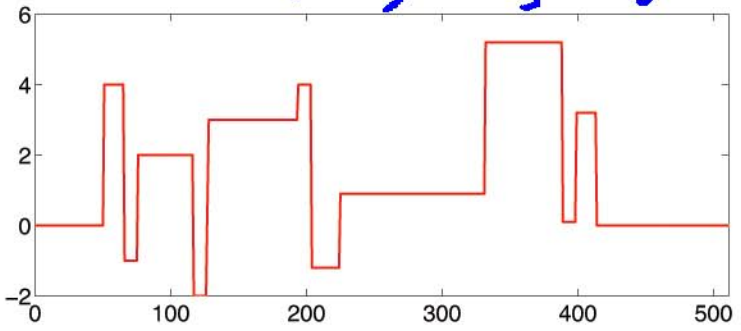
$$P = \left[\begin{array}{ccccc} \text{length} & \text{length} & \text{length} & \text{length} & \text{length} \\ c_{J-3} & d_{J-3} & d_{J-2} & d_{J-1} & x \end{array}\right]$$

IDWT

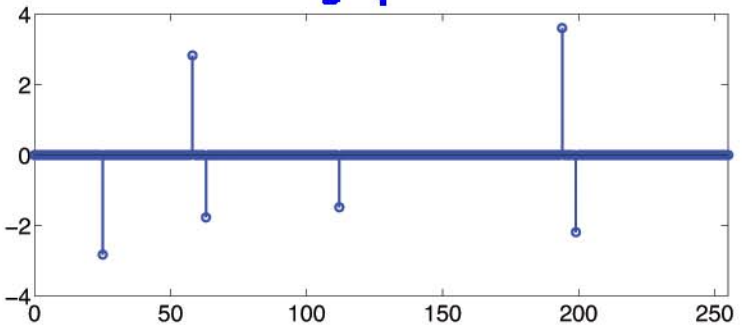
x = waverec(W, P, 'wname')

Example -

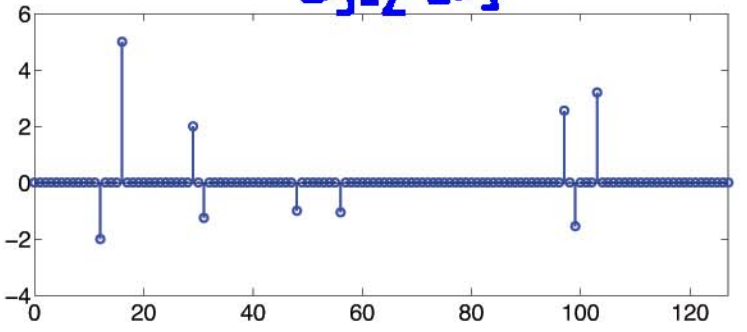
$x(t) \sim c_J[k]$



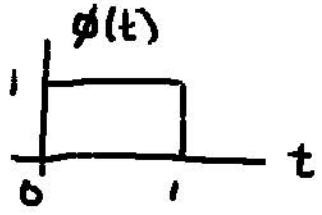
$d_{J-1}[k]$

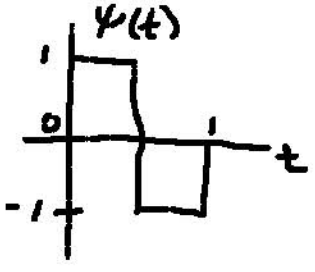


$d_{J-2}[k]$

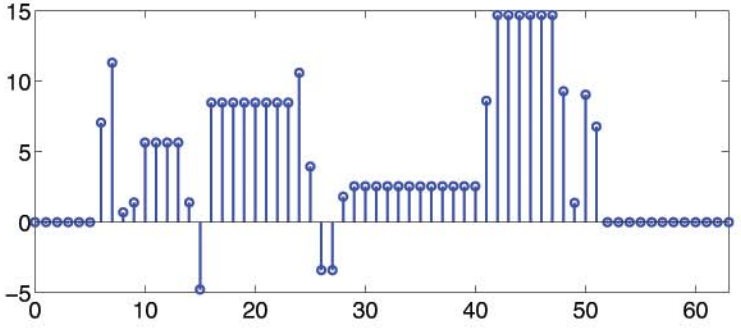


[W, P] = wavedec(blocks, 3, 'haar');

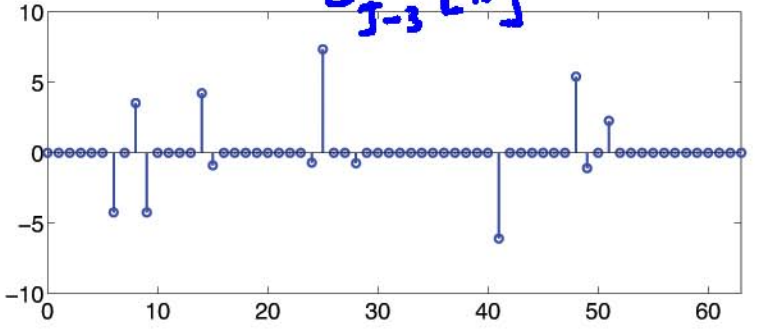




$c_{J-3}[k]$



$d_{J-3}[k]$



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